Independent exact certification of machine-generated mathematics: the first register

Carlos Toledo*

October 2026 — draft v0.1 for review; every number interpolated from the record; machine-derived, not peer-reviewed; author list to be agreed

Abstract

Mathematics now arrives from machines as exact objects with enclosures: constants, counterexamples, packings, matrix-multiplication schemes and certificates for asymptotic bounds, each validated by the checker of the group that produced it. This paper describes a method for deciding such claims independently — in exact rational and outward-rounded interval arithmetic, with no code shared with the claimant, and with refusal as a published result — and presents its evidence, a register of 132 published claims decided between 2026-08-25 and 2026-10-05: 107 certified, 2 refuted, 9 partial, 5 mixed, 6 repaired, 2 needing data the claimant has not published, and 1 queued and not counted. Every row is derived from a record that re-runs, carries the witness its verdict needs, and names what went wrong from a closed vocabulary of 14 defect kinds. Most claims hold: 107 of the 131 decided rows name nothing wrong. The defects that do appear are of the verifier's kind rather than the mathematician's — a floating-point product printed as an exact constant, witnesses that hold only within a verifier's tolerance, a checker that accepts what its own definition excludes — and one row keeps as its history a clause present in a paper's theorem and absent from its formal statement, until the claimant's next commit proved it. The novelty claimed is the decision grammar (verdict, witness, refusal as a result) and the public re-runnable record, not interval arithmetic. The method runs both ways: a theorem of this laboratory re-certified by an outside audit with no shared code is recorded like any other row.

1 Introduction

1.1 Where the claims come from

In 2025 and 2026 a large part of newly announced mathematics has been produced, in whole or in part, by machines, and has arrived in a form that is new: not a proof to be read but an object to be checked. AlphaEvolve [1, 2] publishes packings, step functions and polynomials as coordinate files; AlphaTensor [3] publishes tensor decompositions; agent platforms such as

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

EinsteinArena [5] and the Station [6] publish kissing configurations and tables of constructions scored by their own verifiers; test-time discovery systems [7] publish step functions with six hundred steps. A benchmark, HorizonMath [10], credits frontier models with discoveries and accepts a closed form on a twenty-digit numerical match while saying that this is not a proof. A library of counterexamples found by language models [11] ships a certificate per case. A registry of optimisation constants marks bounds whose verification is "at minimal levels" with an asterisk [19]. A paper on diagonal Ramsey numbers prints, after a remark, an improvement proposed by a chat model and labels it preliminary and unverified [12]. Counterexamples to the Jacobian conjecture in dimension three [13] and to the weak Markus–Yamabe conjecture in dimension fourteen [14] are polynomial maps whose claimed properties are identities. On a problem thread a model-produced constant was posted to thirteen decimals with no error bound [21].

What these have in common is that each claim comes down to finitely many exact arithmetic facts, and that each was validated, if at all, by the checker of the group that produced it. The tools to decide such facts without error are old: interval arithmetic with outward rounding [25, 26], exact rational arithmetic, and the discipline of computer-assisted proof that reached its public standard with the formal Kepler proof [27]. What did not exist was an independent party applying them to the stream, with a grammar for the outcomes and a public record of every decision, including the ones that went nowhere. A verifier that only ever passes or fails is a scorer, and scorers are what the stream is full of; an audit of reward verifiers in reinforcement learning found a reference numeric verifier that accepts every off-by-one answer above ten thousand [24]. The missing third outcome — this claim cannot be decided here, and here is why — is the one a motivated party never builds.

1.2 What this paper is

This is the method paper of a project whose position, adopted on 2026-09-03 and held since [32], is one sentence: *independent exact certification of machine-generated mathematics — exact arithmetic, no code shared with the claimant, refusal as a verdict.* The paper states the decision grammar (§3), the discipline that keeps it honest (§4), and the register that is its evidence (§5): 132 rows derived from 25 records, each decided by an instrument written from the published statement and the published bytes, never from the claimant's code. Three rows are worked in a page each (§6): a refutation with its witness, a repair, and a refusal with the one thing that would decide it. Section 7 reads the register as a measurement of what goes wrong in machine-generated mathematics; §8 records the direction the method must also run in, a theorem of this laboratory re-certified by an outside audit with no shared code. Sections 9 and 10 say what is not claimed and where the method stops.

The project's own mathematical records — the value of $\lambda(4)$ in Chowla's cosine problem [28], the dimension-eleven kissing ledger [30], the explicit sum-distinct sets below Bohman's constant [31] — are cited as records and not restated. The novelty line is deliberately narrow: not interval arithmetic, which is standard, but the grammar of verdict, witness and refusal, and a register every row of which re-runs from a public record.

2 Statement of results

The results are of three kinds, and the paper keeps them apart: counts, which are facts about a record; theorems, which the record proves; and decisions, which are the instrument's outcomes stated as such.

Proposition 1 (the register, as recorded)**.** *As of 2026-10-27 (git aae61990), the register of §5 holds 132 rows, each derived from one of 25 records and dated by the first commit whose record held it. 131 are decided and 1 is queued: 107* CERTIFIED, *2* REFUTED, *9* PARTIAL, *5* MIXED, *6* REPAIRED, *2* NEEDS DATA*. Every row is self-initiated and 0 were submitted by others. 107 decided rows name no defect; 24 name one of 11 kinds from a closed vocabulary of 14.*

Theorem 1 (a published constant, refuted)**.** *Let* $C^* = \frac{1}{2}\big(\prod_{p\geq 3}(1+1/(p-1)^3)-1\big)$*, the product over odd primes. Then*

$$0.0752403861783092455893 \;<\; C^* \;<\; 0.0752403861783095651674,$$

and the value 0.0752403861777 *published for* C^* *on the Erdős problem #852 thread on 2026-04-24 is not* C^* *under either a truncation or a rounding reading: it parts from the enclosure at its 12th decimal place (the 11th significant digit). The refutation needs no tail bound: with* N/D *the exact partial product over primes* $3 \leq p \leq 400{,}000$*, a strict lower bound for the full product, one integer inequality,*

$$5 \cdot (N - D) \cdot 10^{12} > 752403861778 \cdot D,$$

already places $(N/D - 1)/2$ *above the upper edge of the published window.*

Proposition 2 (a registry constant, repaired)**.** *Decided from the note the registry cites [20], given that note's chain (the explicit constant of its source, its lattice-doubling lemma and its regulator bound): for every choice of the chain's parameters* $s > 1$*,* X *and* Y *the exponent satisfies* $c \leq 0.0007150508 < 0.000719$*, so the registry's asterisked bound* $C_{84b} \leq 1.999281$ *(*$c \geq 0.000719$*) is out of reach of the calculation cited; the note's own theorem,* $c \geq 0.0007$ *(*$C_{84b} \leq 1.9993$*), holds, and at the note's own parameter choice* $c = 0.0007062455$*.*

Decision 1 (a printed design table, refused)**.** *The 20 annual-maximum* H_s *fits printed for 5 South Atlantic lease areas [18] are decidable and not decided:* NEEDS DATA*. They are fits to a commercial hindcast (WaveClimate, 1992–2022) that nobody outside can re-run. Against the nearest open-sea cells of the public hindcast the paper also names (Ifremer GLOBMULTI_ERA5_GLOBCUR_01, 1993–2024, CC BY-SA 4.0), 31–61 km from the printed coordinates, 19 are off the certified maximum-likelihood fit and 1 gives the record zero density — which measures the distance between two records, not an error in the table. One file decides the row: the annual maxima the authors fitted.*

Proposition 3 (a theorem of this laboratory, re-certified outside)**.** *The proof of the value of* $\lambda(4)$ *[28] decides 2,231 finite sets across 9 families (6 skips, all of the extremizer). On 2026-09-30 an outside audit with no shared code [29] re-derived the cubic, the 14 generic collisions and the 9 families from the write-up, re-certified every finite case in interval arithmetic, and swept every gcd-reduced* 4*-set with largest element at most 80 as a proof-independent control, finding* $\{1, 2, 3, 4\}$ *deepest and the nearest rival {2,3,4,6} at about* -1.774*. Its three documentation findings were folded into the write-up; no step broke.*

3 The grammar

Definition 1 (certified)**.** A claim is *certified* when a finite set of exact arithmetic facts that imply it has been decided — each in exact rational arithmetic (which decides ties) or in outward-rounded interval arithmetic (which bounds what cannot be computed exactly) — and a certificate re-runs without this engine present and carries the input that would make it fail. "Certified"

is used in this paper in that sense only; a result decided this way is said to be *decided*, and when the facts amount to a proof, *proved*.

The instrument returns three verdicts and no fourth. CERTIFIED and REFUTED are theorems about the claim as stated, each carrying a witness: for a certification, the certificate and its falsifier; for a refutation, the printed falsifying object — an inequality, a pair outside a region, a violated equation. REFUSED is what a claim receives when the instrument cannot decide it: a real outcome, published like the others, and evidence of nothing either way. A refusal is terminal. It is never retried at lower rigour, never converted to a probability, and never dropped from the record; the only path from REFUSED to CERTIFIED is a new exact argument.

The register's other words are compositions of these three, not additional verdicts, and each names which of the three it is composed of (Table 1). The register states it in its own header: CERTIFIED and REFUTED are theorems, PARTIAL names the fragment that was reached, NEEDS DATA measures the claimant rather than the claim, MIXED means the row is an aggregate whose record holds both, and REPAIRED means the claim as printed is refuted and an object next to it is certified.

word	composed of	the witness the row carries
CERTIFIED	—	a certificate with its falsifier; re-runs without the engine
REFUTED	—	the falsifying object, printed
REFUSED	—	the reason, in a sentence; nothing else
PARTIAL	certified + refused	which fragment certified and which core was not reached
NEEDS DATA	refused, cause with the claimant	the bytes that would decide the row, named
MIXED	certified + refuted, one record	the counts of both, and the items refuted
REPAIRED	refuted + certified	the object as printed refuted; the object beside it certified, with its deficit
QUEUED	not a verdict	nothing; the row is not counted as decided

Table 1: The grammar. Three verdicts; four compositions, each of which says what it is made of; and one state that is not a verdict.

Two consequences follow and are enforced rather than recommended. First, a scope sentence is load-bearing: a CERTIFIED without the scope beside it says something the record does not, because several rows certify the computational fragment of a claim whose analytic core was never touched. Second, the kinds of refusal are never summed. The instrument declining (REFUSED), the claimant having published nothing (NEEDS DATA), a budget running out with the attempt recorded (OPEN), a swept cell whose box did not close (UNDECIDED) and an audit stating its reach (SCOPE) are five behaviours; one "total refusals" number would be a bigger number and a smaller fact, so the project's refusals page carries each kind with its own denominator and has no total (Table 9).

4 The discipline

Five rules make the grammar hold under pressure. They are stated as engineering, because each is enforced by code that refuses rather than by a sentence that advises.

Screens prune, never admit. Floating point decides only what is worth certifying. Nothing a screen passes is believed; every admission is an exact decision. In the project's generation loop, the campaign runner behind its own records, a screen is required before any run to recover a planted set of known answers, so a filter that quietly drops hard cases refuses the run by name [33].

Red controls must fire. A check that has never gone red is decoration. The certificate constructor in the interval toolkit refuses to build a certificate without a falsifier — the input that would make it fail — and refuses a PROVED verdict with no evidence. Every battery carries deliberate forgeries that must be rejected: in the six-lane manuscript audit, 21 mutation controls stand beside 92 named checks and a control that stops firing refuses the page; the campaign runner ships 19 ways to cheat its own reward signal, each caught by a named control at every build [33]. The methods note records 16 real defects this project shipped and then found: none was found by reading code; all were caught by an impossible number, a calibration, a control, a byte pin or an outside read (Table 2).

Pin the bytes. Every transcribed claim is decided against a held byte sequence, path and sha256, re-hashed at decision time. One catalogued bug — a row silently missing from a transcription of a published sheet — was caught by nothing else.

No code shared with the claimant. The claimant's program is never in the trust path: the decision is re-derived from the published statement and the published bytes. The precise form, adopted after a check of the unqualified phrase [32], is *independence from the claimant*: two of the project's own instruments reuse the project's own code across a producer/checker line, and the positioning note names both. Where an author's script was read to settle which identities a manuscript meant, the lane says so and the script was not executed.

Honest counting. Every row is derived from a record; a claim with no record gets no row. An aggregate audit — a dataset's keys, a benchmark's counts — is one row that counts once and names its most frequent defect, because a key, a contour and a construction are not units of one another. A correction the project adds beside a refuted row is never counted among the audited corpus's rows. A queued row is not a decided row. Origin is tracked separately, and the register's submitted count stays at 0 until someone sends something. Refusal kinds are never merged. Each of these rules is a line that makes a build refuse.

caught by	defects
code read	6
impossible number	3
outside read	2
calibration	2
control	2
byte pin	1

Table 2: The 16 defects of the methods note [33], by what surfaced each; parsed from the note's own catalogue, which runs 7 batteries at every build and refuses if one goes red.

5 The register

5.1 How a row is made and dated

The register is derived, never typed. The tool that writes it reads 25 records on disk, each the output of an instrument that decided some published claim, and emits one row per claim with the claimant, the source as pinned, the verdict, the scope sentence, the defect kind, and the record and page it came from. A row names its defect kind from one closed vocabulary defined in that tool and nowhere else; a defect that fits no kind refuses the register until the vocabulary is decided on, never "other". The day a row entered is the first commit whose deciding record held the row's key, found with git's pickaxe on the record, so the register is a dated diff of the records rather than a log of sessions. The monthly ledger is that diff cut by month, with its first instance September 2026 [35].

5.2 By source and verdict

Table 3 is the register grouped by the record each row was derived from, in the order the records entered. The claimant column is coarse on purpose: it says whose bytes were decided, not whose mathematics is better.

Three things in the table deserve a sentence each. The 10 rows of the matrix-multiplication record are 2 calibrations (Strassen's scheme [4] and its exact Kronecker square), 7 AlphaTensor schemes and 1 AlphaEvolve scheme; 2 of them are certified over $\mathbb{F}_2$ and refuted over $\mathbb{Q}$, which is not a defect but the content of the claim, and the scope sentence says so. The 20 EinsteinArena table rows decide every construction that platform's state-of-the-art table publishes, across 8 problems and including one human construction [8], from the bytes as the decimals they print: 16 are exact witnesses of the value stated and 4 are witnesses only within a verifier's tolerance (the platform's 10^{-9} on one circle packing, in force when it was published and removed since; the repository notebook's 10^{-6} on three step-function sums), each repaired to an exact witness with its deficit printed. The one QUEUED row of the register is the 2022 kissing record the machine ladder started from [9], whose bytes have not yet been pulled. The 14 rows of the counterexample library [11] were decided by standard-library programs written from each case's statement before the library's checkers were read: 9 certified whole, 5 partial, none refuted.

5.3 The dated diff

Rows entered on 13 days between 2026-08-25 and 2026-10-05 (Figure 1b). The largest day of August and September, 2026-09-29 with 33 rows, is the day the register's schema was extended over every published-claim audit on disk. The split of those two months — 18 rows in August against 69 in September — follows the adoption of the position on 2026-09-03, after which the sources of Table 3 were sought out. October's 45 rows, to 2026-10-05, are of another making. All but one are decided rows of the 100 machine claims pre-registered on 2026-10-02 before any was decided [23]: 44 rows from 5 of its 10 pools so far, among them the largest day of the register, 2026-10-04 with 35 rows. The other is the Navier–Stokes row, re-decided on its claimant's later commit and dated by the record that decided the new verdict (§7).

record	whose bytes	rows	CERT	PART	REF	MIX	REP	ND	Q	entered
Erdős #852, the constant C^*	a problem-thread post, frontier-model help	1	0	0	1	0	0	0	0	2026-08-25
matrix-multiplication schemes	Strassen, AlphaTensor, AlphaEvolve	10	10	0	0	0	0	0	0	2026-08-25
Ramanujan Machine result sheets	the Ramanujan Machine project	1	0	0	0	1	0	0	0	2026-08-25
six AI-assisted manuscripts, lane audit	manuscripts produced with frontier-model help	6	5	1	0	0	0	0	0	2026-08-30
kissing configurations in $\mathbb{R}^{11}$	AlphaEvolve, EinsteinArena, the Station, classical	11	10	0	0	0	0	0	1	2026-09-03
EinsteinArena state-of-the-art table	AlphaEvolve, TTT-Discover, Together AI agents, Haugland	20	16	0	0	0	4	0	0	2026-09-08
environmental-contour benchmark counts	the benchmark's teams	1	0	0	0	1	0	0	0	2026-09-15
GSM8K answer keys	GSM8K	1	0	0	0	1	0	0	0	2026-09-21
METR time-horizon fits	METR	1	0	0	0	1	0	0	0	2026-09-22
printed H_s marginal fits	the benchmark's teams	1	0	0	0	1	0	0	0	2026-09-27
a printed design table	Bhaskaran et al.	1	0	0	0	0	0	1	0	2026-09-28
an AI counterexample library	Sra; GPT, Codex and Claude models	14	9	5	0	0	0	0	0	2026-09-29
registry asterisk C71	Numaro	1	1	0	0	0	0	0	0	2026-09-29
GNNW's unverified iteration	ChatGPT 5.6 Sol, printed by Gupta et al.	1	1	0	0	0	0	0	0	2026-09-29
HorizonMath credited discoveries	GPT-5.4 and GPT-5.6 models	3	1	0	1	0	0	1	0	2026-09-29
polynomial maps, two 2026 papers	Gao with Claude; Castañeda et al.	8	7	1	0	0	0	0	0	2026-09-29
registry asterisks C3b, C3c	Mosaic Intelligence, Y. Lin	3	3	0	0	0	0	0	0	2026-09-29
registry asterisk C84b	Althoefer with ChatGPT	1	0	0	0	0	1	0	0	2026-09-29
registry asterisk C42	Griego; Röhrig with Codex	2	0	2	0	0	0	0	0	2026-09-29
AlphaEvolve notebook, pre-registered	AlphaEvolve	15	15	0	0	0	0	0	0	2026-10-04
AlphaTensor over $\mathbb{F}_2$, pre-registered	AlphaTensor	8	8	0	0	0	0	0	0	2026-10-04
AlphaTensor, standard arithmetic, pre-registered	AlphaTensor	12	12	0	0	0	0	0	0	2026-10-04
EinsteinArena bests, pre-registered	EinsteinArena agents	8	7	0	0	0	1	0	0	2026-10-05
the Station, pre-registered	the Station's agents	1	1	0	0	0	0	0	0	2026-10-05
Navier–Stokes Lean certificate	OpenAI	1	1	0	0	0	0	0	0	2026-10-05

Table 3: The register by source. 132 rows from 25 records on 22 pages; a row counts once, an aggregate row is one audit. The entered column is the first commit whose record held the source's first row.

month	rows	CERT	PART	REF	MIX	REP	ND	Q	with a defect
August 2026	18	15	1	1	1	0	0	0	3
September 2026	69	48	8	1	4	5	2	1	20
October 2026	45	44	0	0	0	1	0	0	1

Table 4: The register by month, as the monthly ledger cuts it. A row belongs to the month in which its deciding record first held it.

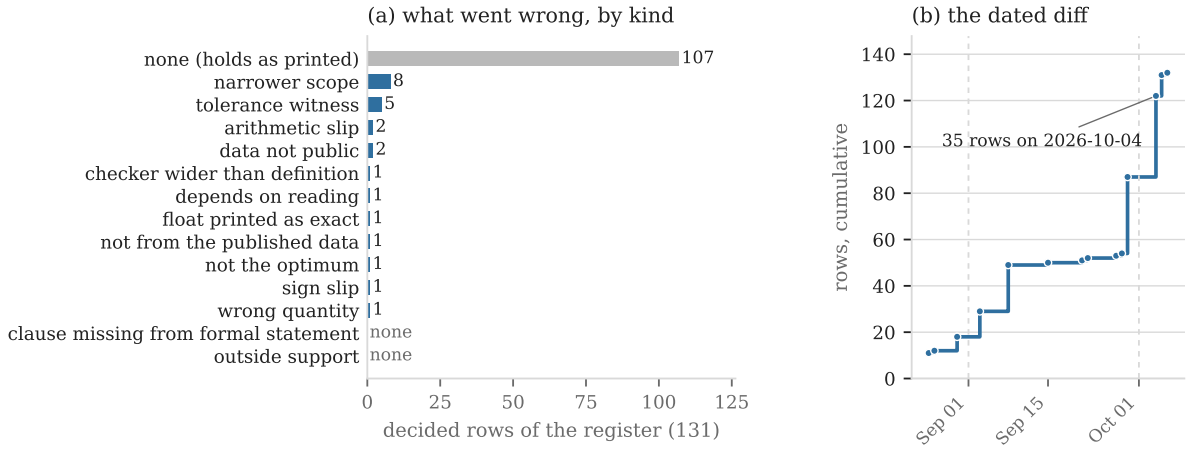


Figure 1: (a) The 131 decided rows by defect kind; the vocabulary's kinds with no row are drawn empty. (b) The register as a dated diff: cumulative rows by the day a record first held them, with the month boundaries dashed. Both panels are drawn from `certs/claims-ledger.json`.

6 Three worked examples

6.1 A refutation and its witness

Erdős problem #852 concerns a counting function whose conjectured asymptotic rests on two constants; both were posted to the problem thread on 2026-04-24 as bare decimals, produced with a frontier model's help [21]. The second, C^*, was printed to 12 significant digits, 0.0752403861777.

The instrument encloses C^* by a directed-rounding partial product over the 1,857,858 odd primes below 30,000,000 at 192 bits, with the tail bounded by $\sum_{p>L} 1/(p-1)^3 \le 1/(2(L-1)^2)$ and $\prod(1+a_p) \le e^S \le 1+S+S^2$; the enclosure is the one in Theorem 1. The published decimal lies below it. That alone is a refutation, but it rests on the tail bound and on the rounding mode, and a reader who distrusts either should not have to. So the certificate also carries a witness that needs neither: the exact rational $N/D = \prod_{3\le p\le 400{,}000}((p-1)^3+1)/(p-1)^3$, a strict lower bound of the full product because every further factor exceeds one, already satisfies $(N/D-1)/2 > 752403861778/10^{13}$, the upper edge of the published claim's window under both the truncation and the rounding reading. That is the one integer inequality of Theorem 1, and a standard-library verifier (`tools/verify_erdos852.py`) re-decides it with no code from this repository.

The record also names the mechanism, which is the point of the example for the register. The published decimal equals the naive IEEE-754 double product of the defining formula: the factor $1 + 1/(p-1)^3$ rounds to exactly 1.0 once $(p-1)^3 \ge 2^{53}$, that is for $p - 1 \ge 208{,}064$, so about 87% of the factors vanish and the floating-point product truncates itself near $p \sim 2 \cdot 10^5$ whatever the loop bound. The register's kind for the row is *float printed as exact*. The thread's other constant, $c_0 = 1.32322827686395\ldots$, is correct as a rounding and is recorded so, with 61 certified decimals beside it: a grammar that can say "correct as a rounding" keeps a refutation from reading as a judgement of the whole post.

6.2 A repair

The optimisation-constants registry [19] (commit 2c1968cd) marks bounds whose verification is at minimal levels with an asterisk; its row for the real sum–product exponent quotes, from a note of 2026-05-28 by Ingo Althoefer written with a chat model [20], the bound $C_{84b} \leq 1.999281$, that is $c \geq 0.000719$, and marks it unverified. The note's own theorem claims only $c \geq 0.0007$ and says that the optimised value suggested by the same calculation is about 0.000719, with 0.0007 the safer constant to quote. The registry quoted the suggestion.

The instrument decided the note's chain: its seven numerical facts, each in interval arithmetic, and then the one question the registry row turns on — whether any choice of the chain's free parameters reaches the quoted exponent (Table 5). None does: the chain's ceiling is $c \leq 0.0007150508$, below 0.000719, and the note's own choice gives $c = 0.0007062455$. The row is therefore REPAIRED: the claim as the registry prints it is refuted, and the object next to it — the note's theorem, $c \geq 0.0007$, given its cited inputs — is certified. The kind is *arithmetic slip*: a printed arithmetic step that does not hold, here the passage from a calculation to a constant the calculation cannot produce. What the row does not decide is the chain's inputs, which are a published paper's and a lemma's; Proposition 2 is conditional on them and says so.

	the check, as the record names it	what it found
1	`f(1.371966384) <= 103 (the note: about 101.956)`	`[101.9560758327, 101.9560758328]`
2	`515/Y = 515/1415 < 0.364`	—
3	`M1 e^Y/(X Y^2) + M2/(X^2 Y^2) < 0.136 (and so < 0.364)`	`0.1352394770`
4	`log|A|/d < L < 1432`	`1431.1139758`
5	`-log(0.364) > 0.0007 * 1432 = 1.0024: c = 0.0007 holds`	`1.01060141`
6	`inf over s > 1 of f(s) >= 101.9 (a sweep of [1.01, 2.9] by 0.001, of [1.2, 1.6] by 0.0001; below, zeta > 100; above, the power dominates)`	`smallest endpoint bound 101.927484 at s = 1.372`
7	`for every s > 1, X and Y, the chain gives c <= 0.0007150508 < 0.000719`	`the maximum near Y = 1399.0`

Table 5: The 7 checks of the C84b record, verbatim. The last is the one that repairs the row.

The same composition decides a different failure in the EinsteinArena table. A circle packing published as $\Sigma r = 2.3658323759$ has, in exact arithmetic on its own bytes, 44 overlapping pairs and a bounding box over its limit — all within the 10^{-9} slack the platform's verifier allowed when the packing was published (2026-04-02), a slack the platform removed on 2026-08-24; its live verifier, read on 2026-10-06, rejects the file. Scaling every radius by the largest rational λ that clears every pair and the box gives an exact witness of $2.3658323758\ldots$, a deficit of 8.67×10^{-11} against the printed value; the printed digits are not the exact value's, and the row says which digit moved. The 3 minimum-overlap step functions whose sums miss $n/2$ by up to 1.48×10^{-14} are repaired the same way, by a rescaling that moves each bound by at most 8.32×10^{-18}. In every repaired row the bound survives; what the repair records is that the bytes as published were not yet a witness, and by how much.

6.3 A refusal, and what would decide it

A published design table [18] prints annual-maximum H_s fits — four methods, 5 South Atlantic lease areas, 20 rows — from a commercial hindcast (WaveClimate, 1992–2022) that cannot be re-run outside it. The claim is of the decidable kind: a fitted distribution either is or is not the likelihood's maximum on the series it was fitted to, and the project's return-level instrument

decides that by a Krawczyk box on the score with the Hessian proved negative definite over it, as it has on the public benchmark buoys [17]. What is missing is the series.

The row is NEEDS DATA, kind *data not public*. It is not left empty. Against the nearest open-sea cells of the public hindcast the paper also names (Ifremer GLOBMULTI_ERA5_GLOBCUR_01, 1993–2024, CC BY-SA 4.0), 31–61 km from the printed coordinates, with 32 calendar-year maxima each, the printed fits are 19 times off the certified maximum and once outside the record's support (Table 6). The instrument reports these as the distance between two records — two hindcasts at two points — and refuses to call them errors in the table, because the verdict would depend on data the authors have and the instrument does not. What would decide the row is one file: the annual maxima the authors fitted at the five sites. With it, each printed row decides in minutes, as the printed fits of the contour benchmark's teams did.

lease area	Gumbel, least squares	Gumbel, max. likelihood	Gumbel, moments	GEV, max. likelihood
UY01	10.09 / 8.38	9.48 / 8.38	9.70 / 8.38	9.01 / 9.61
Projeto Açu	5.55 / 3.21	5.46 / 3.21	5.33 / 3.21	6.36 / 3.22
Farol wind	7.84 / 5.86	7.49 / 5.86	7.54 / 5.86	7.40 / 5.76
Sopros do RJ	4.06 / 5.47	3.98 / 5.47	3.94 / 5.47	4.14 / 5.04
Projeto Ubu	3.20 / 4.93	3.03 / 4.93	3.07 / 4.93	2.83 / 4.86

Table 6: The 20 printed 100-year H_s levels (m) against the certified fit's at the nearest public cell, as printed / certified. The differences, from 0.59 to 3.14 m, measure two records' distance, not the table's correctness; a NEEDS DATA row publishes them so the reader can see what the missing bytes would settle.

The precedent for this kind of row is in the kissing ledger [30]. On 2026-09-03 the headline 604-point configuration of one platform was not published; the row read NEEDS DATA and named the bytes that would decide it. The platform published them on request on 2026-09-07, 4 days later, and the row decided CERTIFIED in 65 ms with 19,704 exact contacts. NEEDS DATA is the one verdict that can go to zero by the claimant's act, and the register is built so that the act is cheap.

7 What the register measures

A register of self-chosen claims is not a sample of the field (§9). What it measures is the *kind* of thing that goes wrong, and the vocabulary was built for that: closed, each kind defined in one sentence, a new kind a decision about the world made in one place (Table 7).

Four readings of the table are supported by the record.

Most claims hold. 107 of the 131 decided rows name nothing wrong, 82%; the 109 rows that are theorems (CERTIFIED or REFUTED) split 107 to 2. Inside the aggregates the proportion is higher still: of the GSM8K keys [15], 27,996 of 27,998 calculator annotations are exact (2 were refused) and 26 printed prose steps of 8,792 keys do not hold; of the contour benchmark's 150 contours, 174 of 176 printed exceedance counts agree with the exact count (173 exactly) and 2 do not; of METR's 44 time-horizon fits [16], all are certified and the printed coefficients are the rounding of the certified optimum for 22 of 23 models. A register that led with its refutations would misreport its own content.

The most frequent defect is a statement about reach, not error. Narrower scope (8 rows) is the kind of all but one PARTIAL row (that one depends on a reading only its author can

kind	definition	rows	items
none	the claim holds as printed	108	359
narrower scope	decided only in a narrower scope than printed; the rest is out of reach, not wrong	8	8
float printed as exact	a floating-point result printed as the exact quantity	1	1
sign slip	a sign wrong in a printed constant	1	1
arithmetic slip	a printed arithmetic step that does not hold	2	27
tolerance witness	a witness only within a numerical tolerance; the printed value is the tolerance's	5	5
not the optimum	printed as an optimum, but not the one the definition names (a stop short of it, or a local one)	1	1
wrong quantity	the printed number is a correct answer to a different question	1	1
not from the published data	the printed number is not what the published data give, and the arithmetic is not why	1	2
outside support	the printed model gives observed data zero density	0	0
clause missing from formal statement	a clause the prose claims is absent from the formal statement that is proved	0	0
data not public	the data behind the claim are not public, so no one outside can decide it	2	2
depends on reading	the claim holds under one of two definitions its own text gives, and not under the other	1	1
checker wider than definition	the checker that accepted the claim admits what the definition it encodes excludes	1	1

Table 7: The closed vocabulary of 14 kinds. Rows: register rows of that kind (an aggregate row by its most frequent defect; the queued row counts under none). Items: the same with the 5 aggregate rows opened, so that the 283 keys, contours, fits and sheet rows inside them are counted one by one.

settle): a certified fragment beside a core — an asymptotic reduction, an all-m argument, a cited realisation theorem — that is prose and was not touched. The six-lane audit of AI-assisted manuscripts is the clearest case (Table 8): 5 of 6 lanes certified and 1 partial, and in all 6 the computational fragment certified while the analytic core did not, because the core is not machine-checkable at any budget this project has.

The defects are the verifier's, not the mathematician's. Of the 11 kinds that occur, the ones that name a wrong mathematical object are rare: one sign slip in a printed sheet, one float printed as exact, two arithmetic slips (one of them the 26 GSM8K steps). The others describe the checking: witnesses that hold only within a verifier's tolerance (5 rows); a checker that accepts a pair if either of two one-sided tests passes where its definition needs both (1 row, §7.1); a claim that holds under one of two definitions its own text gives (1 row); a printed number that is a correct answer to a different question (1 row); printed numbers not what the published data give (1 row); an optimiser's stop printed as the optimum (1 row); and data nobody outside can decide on (2 rows). One kind occurred and has left the register: a clause in a paper's theorem that is absent from the formal statement that was proved. Its one row, OpenAI's Navier–Stokes certificate [22], read PARTIAL from 2026-09-09 at the announced commit 8937a8f4: 2,486 modules built here with only the standard axioms, and the finite-energy clause of the paper's Theorem 1.1 not in the statement. The one upstream commit since, f9e8bc5b of 2026-09-10, proves that theorem with the clause; rebuilt here on the standard axioms, it moved the row to CERTIFIED on 2026-10-05, with the PARTIAL kept as the row's history and the kind left with 0 rows (Table 7). The defect was real at the commit the claim was announced on, and the register keeps both dates.

Who produced the bytes is not what this register measures. By a coarse rule — the claimant

field names an AI system — 107 decided rows are machine-produced and 24 are not; 17 of the former and 7 of the latter name a defect. The rule is a regular expression over a text field and the register was not sampled; the number is reported because a reader will ask. Here the machine-produced rows name a defect less often, but a coarse rule over an unsampled register supports no conclusion about either population.

lane	verdict	scope	checks	controls
Maxwell's point-charge bound	CERTIFIED	at $\varepsilon = 1/6$ only	17	3
The Korenblum constant	CERTIFIED	the numerical criterion	15	4
Erdős Problem #1038	CERTIFIED	the computational fragment	22	4
Ran–Teng Conjecture 20	PARTIAL	machine-checkable fragment only	38	5
The Mathieu property for Lie groups	CERTIFIED	supporting identities only	—	3
The rank-two Poisson conjecture	CERTIFIED	the explicit counterexample	—	2

Table 8: The six-lane audit [36]: six results produced with frontier-model help, re-verified from the manuscripts by verifiers that never executed the authors' code; 92 named checks from the 4 lanes whose verifiers print a ledger, 21 mutation controls, 4.6 s at every build. The lane pages say CONFIRMED where the register says CERTIFIED. A verifier that prints no per-check ledger is shown with a dash rather than an invented count.

7.1 The second refutation: a checker wider than its definition

HorizonMath [10] credits a frontier model with a certificate improving the Gupta–Ndiaye–Norin–Wei bound on diagonal Ramsey numbers [12] from 3.7992 to $3.6960839126\ldots$, validated by the benchmark's own checker. The instrument re-derived the constant and then decided the one thing the checker had not: whether the certificate's pairs lie in the region R that the paper's Theorem 14 requires. The pair at the point where the constant is read, $(X(1), Y(1)) = (0.22745, 0.9988)$, is outside R — Erdős's 1947 bound at $\varepsilon = 13/2000$, $p = 11981/500000$ exceeds the pair's rate by 0.00130 — and 56 of the certificate's 201 decided points are outside R, with 128 more not placed in R by the bound the problem names. The cause is in the problem's own rule: it accepts every pair with $x \le e^{-U(1)} = 0.26292$ whatever y is, asking one of the two inequalities membership needs; the benchmark's validator, at line 299, takes a minimum where the definition takes a maximum, and with that one word changed it fails the certificate on its first interval. The row is REFUTED, kind *checker wider than definition*; what is refuted is the certificate, not the inequality, and the scope sentence says so. The same benchmark's Kakeya construction is CERTIFIED exactly: the union of its 4,033 pieces has area $6008623/55{,}050{,}240 = 0.1091479891\ldots$, below the baseline it claims to beat. The GNNW iteration that the paper itself prints as preliminary and unverified is certified too, by two independent implementations, with $c = 3.78232877553\ldots$, of which the printed 3.78233 is its rounding [36].

7.2 Refusals, by kind, each with its denominator

The refusal with the largest population is neither the instrument's nor the claimant's: 18,001 of 41,367 cells of two exhaustive parameter sweeps (43.5%) did not close at the budget, and are drawn on the atlas as holes rather than filled in by interpolation. A map with no holes drawn is a map that interpolated somewhere.

kind	where	count	of	denominator
REFUSED	the engine loop (the instrument declined to decide an object it enumerated)	73	16,951	objects certified
REFUSED	the matrix-multiplication eval board (a submitted reply carried no parseable proposal)	39	301	claims submitted
NEEDS DATA	the kissing ledger (decidable, but the claimant had published no bytes)	0	11	ledger rows
OPEN	the Erdős #1038 supremum campaign (a degree attempted, budget exhausted, recorded)	1	7	degrees attempted
OPEN	the $\lambda(6)$ campaign (a family still computing, published as unfinished)	1	10	families in the worklist
UNDECIDED	the two-population regime map (a swept cell whose box did not close)	9,915	21,567	cells swept
UNDECIDED	the one-population regime map	8,086	19,800	cells swept
SCOPE	the six-lane audit (a lane that reached a fragment and says which)	6	6	lanes audited

Table 9: Every refusal on record, by kind, as the refusals page keeps them [34]: 5 kinds, each a fraction of its own population, no total. The grader's refusal rate on submitted model proposals is 13.0% (39 of 301; 115 certified, 147 refuted); the 23 replies in which the model declined and the 40 cut off by the harness's own output cap are in neither numerator nor denominator.

8 Records re-certify

A method that decides other people's claims from their bytes must accept the same from others, and the register would be a one-way instrument if nothing ever ran in the other direction. One row has. The proof of the value of $\lambda(4)$ [28] is a finite front: 14 generic collisions and 9 families, within which 2,231 finite sets are decided by certificates against the target enclosure, with the extremizer $\{1, 2, 3, 4\}$ skipped as the definitional witness 6 times, once per finite part it lies on. The project's own audit of it (2026-09-01) walked all 25,819 gcd-reduced 4-sets with largest element at most 30 with code sharing nothing with the engine, re-certified 423 finite-clause sets afresh and found 0 refuters.

On 2026-09-30 an outside audit [29], posted on the Erdős problems issue for the problem, rederived the cubic, the collisions and the families from the write-up with its own code, re-certified every finite case in interval arithmetic, and swept every gcd-reduced 4-set with largest element at most 80 as a proof-independent control; $\{1, 2, 3, 4\}$ was deepest and the nearest rival {2,3,4,6} at about -1.774. Its three documentation findings — a skip count the write-up had stated wrongly, a cone omitted from one family's listing, and a cubic described in the pull request to the problem's site as having a unique root where it has three real ones with $\lambda(4)$ the largest — were folded into the write-up, and the skip count is now computed from the record so the build refuses if it moves. The audit's verdict was that nothing breaks; the register treats it as it treats a row it decided: a record, a date, a witness (the audit repository), and a defect kind for the write-up's prose rather than the theorem.

9 What is not claimed

- No theorem of anyone else's is proved or disproved here beyond its computational fragment. Where a row is PARTIAL, the analytic core was not audited, because it is not machine-checkable; confirming the identities a proof uses is not confirming the proof.
- The register is not a sample. Every row is self-initiated and 0 were submitted by others;

the fraction of rows with a defect is a fact about this register and supports no estimate of an error rate in the field.

- A refutation is of the object as printed. The HorizonMath row refutes a certificate, not the Ramsey inequality; the EinsteinArena repairs refute bytes as witnesses, not bounds; the C84b row refutes a quoted constant, not the note's theorem.
- "No shared code" is independence from the claimant, not from the project's own past. Two instruments reuse the project's own code across a producer/checker line and are disclosed in the positioning note [32]; in one lane an author's script was read, not run, and the lane says so.
- Several claims had already been checked by their authors, in two cases by an interval certificate the authors shipped; an independent reimplementation is not a first verification, and no row says it is.
- Rows conditional on a cited argument are conditional: the C84b repair on the note's chain, the GNNW rows on the paper's Theorem 14, Lemma 15 and Theorem 1, five counterexample-library rows on the universal theorems or all-m arguments their cases cite.
- The register counts rows, not items. The 5 aggregate rows hold 283 items between them; a table that counted those as rows would be a bigger number and a smaller fact.
- One erratum, found while writing this paper. The Ramanujan Machine audit holds 52 certifications: 51 printed sheet rows, of which 50 survive and 1 is refuted as printed, plus 1 certified correction of the project's own, counted separately on the audit page [36]. The register's row for it counts the correction among the printed rows (its text says 52 printed rows), against the project's own counting rule. Verdict, kind and every count above are unaffected; the sentence is to be fixed where the row is derived.

10 Limits

The method's reach is the register's scope sentence: claims that come down to finitely many exact arithmetic facts — exhibit a witness, verify an identity, bound a quantity, decide a constant. Everything else is REFUSED as out of scope rather than guessed at. The instrument cannot decide asymptotics, statements over infinite families with no finite reduction, or proofs whose content is prose; it refuses them, and a refusal says nothing about the claim. Within reach, cost is the limit that shows in the record: a sweep's undecided cells, a campaign's exhausted degree, a family still computing. The vocabulary of defect kinds is closed but not finished; it has 14 entries after 132 rows and will be extended by the first claim that fits none, at which point the register refuses until the kind is decided. Dating by the first commit is exact for the record and late for the work. The register is one laboratory's, and the direction of §8 has run once; that records re-certify is, so far, one row's worth of evidence. Finally, the discipline catches bugs by running the machine, not by reading it (Table 2), which is a reason to trust the controls and not a reason to trust the code.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The register is `certs/claims-ledger.json`, written by `node tools/run-claims-ledger.js` from the records it names in each row's `decidedFrom`; the claims desk, the refusals page and the monthly ledger are built from it by the three `tools/build-report-*.js` builders of those names. The three worked examples re-run from their records: `python3 tools/verify_erdos852.py` decides Theorem 1 with the Python standard library alone; `python3 tools/run-sumproduct-ledger.py --check` and `instruments/sumproduct/battery.py` re-derive Proposition 2; the design-table audit is `certs/design-table-audit.json` with the printed-fit decider `playground/return-level-check/printed.js`. The in-house $\lambda(4)$ audit is `tools/audit-lambda4.js` (record `certs/lambda4-audit.json`); the outside audit is at `https://github.com/rainrzk/erdos510-lambda4-audit`. The macros of this document are written by `tools/paper-numbers/register.js` from the records named in its header, and the figure by `tools/paper-numbers/register-fig.py`; the tool refuses when a record no longer says what a sentence needs. Built at git aae61990d737 from the register of 2026-10-27 (git aae61990).

References

[1] A. Novikov, N. Vũ, M. Eisenberger et al., AlphaEvolve: a coding agent for scientific and algorithmic discovery, arXiv:2506.13131 (2025).

[2] B. Georgiev, J. Gómez-Serrano, T. Tao, A. Z. Wagner, Mathematical exploration and discovery at scale, arXiv:2511.02864 (2025).

[3] A. Fawzi et al., Discovering faster matrix multiplication algorithms with reinforcement learning, Nature 610 (2022), 47–53.

[4] V. Strassen, Gaussian elimination is not optimal, Numer. Math. 13 (1969), 354–356.

[5] F. Bianchi, Y. Kwon, A. Pappu, J. Zou, Harnessing the collective intelligence of AI agents in the wild for new discoveries, arXiv:2606.10402 (2026).

[6] S. Chung, W. Du, W. J. Wesley et al., Autonomous mathematical discovery in an open-world multi-agent environment, arXiv:2608.23691 (2026).

[7] M. Yuksekgonul, D. Koceja, X. Li et al., Learning to discover at test time, arXiv:2601.16175 (2026).

[8] J. K. Haugland, The minimum overlap problem revisited, arXiv:1609.08000 (2016).

[9] M. Ganzhinov, Highly symmetric lines, arXiv:2207.08266 (2022).

[10] E. Y. Wang, S. R. Motwani, J. V. Roggeveen et al., Measuring progress in reasoning toward mathematical discovery with automatic verification, arXiv:2603.15617 (2026).

[11] S. Sra, GPT, the counterexample machine, arXiv:2608.29595 (2026); library at `https://github.com/suvrit/countex-machina`, commit dbf67374 as recorded in `certs/countex-ledger.json`.

[12] P. Gupta, N. Ndiaye, S. Norin, L. Wei, Optimizing the CGMS upper bound on Ramsey numbers, arXiv:2407.19026 (2024).

[13] S. Gao, Counterexamples to the Jacobian conjecture in dimensions greater than two, arXiv:2608.00222 (2026).

[14] Á. Castañeda, G. Honorato, F. Valenzuela-Henríquez, The weak Markus–Yamabe conjecture fails in dimension 14, arXiv:2608.05392 (2026).

[15] K. Cobbe, V. Kosaraju, M. Bavarian et al., Training verifiers to solve math word problems, arXiv:2110.14168 (2021).

[16] T. Kwa, B. West, J. Becker et al., Measuring AI ability to complete long software tasks, arXiv:2503.14499 (2025).

[17] A. F. Haselsteiner et al., A benchmarking exercise for environmental contours, Ocean Engineering 236 (2021), 109504.

[18] S. K. Bhaskaran, A. S. Verma, A. J. Goupee, S. Bhattacharya, A. R. Nejad, W. Shi, Comparison of extreme wind and waves using different statistical methods in 40 offshore wind energy lease areas worldwide, Energies 16 (2023), 6935, doi:10.3390/en16196935.

[19] The optimization-problems registry, `https://github.com/teorth/optimizationproblems`, commit 2c1968cd as recorded in `certs/sumdiff-ledger.json`; its README: “”Bounds for which the level of available verification is currently at minimal levels will be marked with an asterisk””.

[20] Ingo Althoefer, Improved constant for [BSSZ2026], note of 2026-05-28, `https://althofer.de/improved_constant_052.tex`, pinned by sha256 in `certs/sumproduct-ledger.json`.

[21] Erdős problem #852 and its thread, `https://www.erdosproblems.com/852`, pinned by sha256 in `certs/erdos852-certificate.json`; report `https://carlostoledo.co/reports/erdos852.html`.

[22] OpenAI, NavierStokesAndEuler, `https://github.com/openai/NavierStokesAndEuler`, commit 8937a8f4 as recorded in `corpus/navier-stokes/build.json` and commit f9e8bc5b as recorded in `corpus/navier-stokes/upstream-f9e8bc5b.json`; report `https://carlostoledo.co/reports/navier-stokes.html`.

[23] One hundred machine claims, pre-registered 2026-10-02 before any was decided, `corpus/machine-claims-100.json`, decided pool by pool into `certs/mc100-*.json`, `https://github.com/carlostoledo1891/cert-machine`.

[24] E. Xin, Where the verifier fails: a category-level audit of reward signals in RLVR, arXiv:2609.01354 (2026).

[25] R. E. Moore, R. B. Kearfott, M. J. Cloud, Introduction to Interval Analysis, SIAM, Philadelphia, 2009.

[26] W. Tucker, Validated Numerics: A Short Introduction to Rigorous Computations, Princeton University Press, 2011.

[27] T. Hales et al., A formal proof of the Kepler conjecture, Forum Math. Pi 5 (2017), e2.

[28] C. Toledo, The value of $\lambda(4)$: a machine-derived proof completing Mercer's Section-5 program, `https://carlostoledo.co/reports/lambda4.html`, doi:10.5281/zenodo.22225861 (2026).

[29] rainrzk, erdos510-lambda4-audit, `https://github.com/rainrzk/erdos510-lambda4-audit`, posted on `https://github.com/teorth/erdosproblems/issues/392` (2026-09-30).

[30] C. Toledo, Two platforms, one configuration: independent exact certification of the dimension-eleven kissing records, `https://carlostoledo.co/reports/kissing.html` (2026).

[31] C. Toledo, Explicit sum-distinct sets below Bohman's constant: the Erdős problem #1 disproof made effective, certificates at doi:10.5281/zenodo.22800699 (2026).

[32] Positioning: the decisions, `notes/positioning-decisions-2026-09-03.md`, `https://github.com/carlostoledo1891/cert-machine`.

[33] How the bugs were found: the methods note (formerly ”None by reading code”; on 2026-10-09 an adversarial read of the code found six, counted apart), `https://carlostoledo.co/reports/methods-note.html`.

[34] What the machine would not decide, `https://carlostoledo.co/reports/refusals.html`; the claims desk, `https://carlostoledo.co/reports/claims.html`.

[35] Decided, September 2026, `https://carlostoledo.co/reports/decided-2026-09.html`.

[36] Report pages under `https://carlostoledo.co/reports/`: the six-lane audit, `ai-claims-audit.html` (one page per lane, `claim-`*`id`*`.html`); `diagonal-ramsey.html`; `horizonmath.html`; the Ramanujan Machine's sheets, `rm-audit.html`.